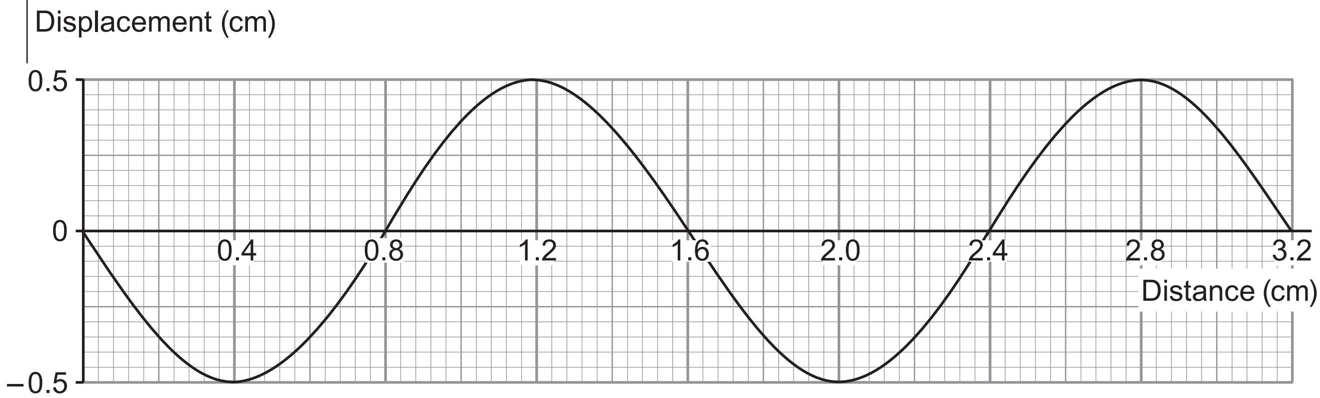
Examiner only

Answer **all** *questions.*

1. A teacher demonstrates water waves in a ripple tank. The diagram represents the waves.



(a) (i) State the wavelength of the wave. .... cm [1]

(ii) State the amplitude of the wave. .... cm [1]

(b) It is observed that 10 waves pass a point in the tank in 2 s.

(i) Underline the correct value for the frequency of the waves.

2 Hz 5 Hz 10 Hz 20 Hz [1]

(ii) Select an equation from page 2 and use it to calculate the wave speed. [3]

Wave speed = ... cm/s

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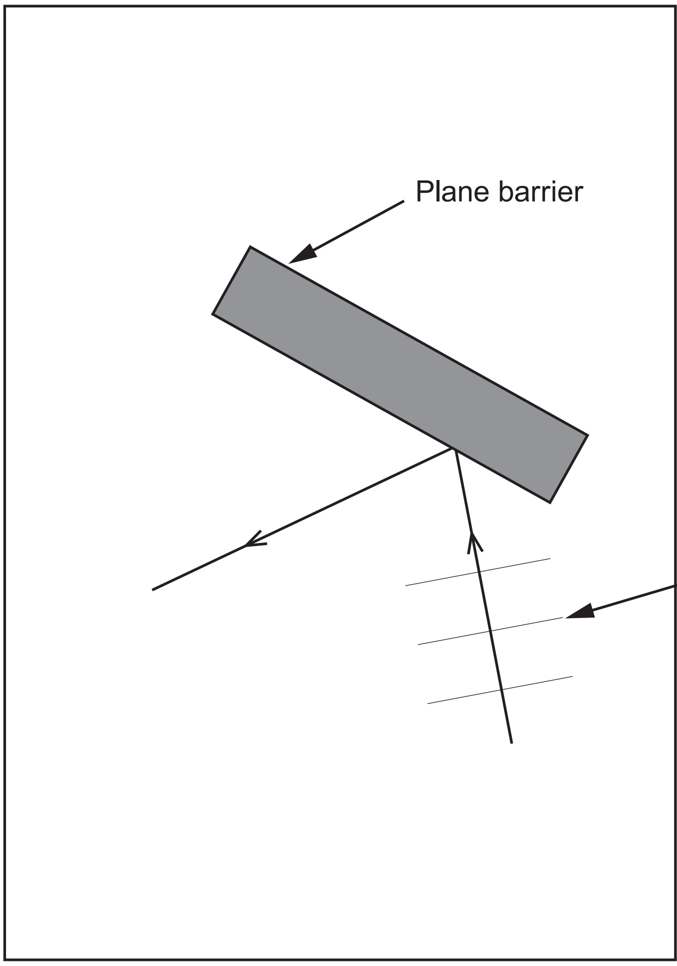
(c) The waves are made to reflect off a plane barrier. **Complete the diagram** to show the reflected wavefronts. [2]



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